

Auctions with Known Asymmetric Values

Auctions as Static Games with Complete Information

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This talk

Games with Pure Strategies

- definitions: game, pure strategy
- static games with complete information
- solution concept: Nash equilibrium

Discrete Sealed-Bid Auctions with Known Asymmetric Values

- sealed-bid first-price auction (FPA)
- sealed-bid second-price auction (SPA)

Cognitive Limits in Auctions

- Texas electricity market (Hortaçsu et al., 2019)

Kick off

Example (Two bidders with known asymmetric valuations, discrete bids)

- $v_1 = 10, v_2 = 5$
- discrete bids $b_i \in \{0, 1, 2, \dots\}$, $i = 1, 2$
- sealed-bid first-price auction (FPA)
- if $b_i > b_{-i}$, the bidder i wins (win/loss)
- if $b_1 = b_2$, each bidder wins with probability $\frac{1}{2}$ (tie/tie)
- in FPA, the price is $\max\{b_1, b_2\}$
- in FPA, the winner's profit is $v_i - \max\{b_1, b_2\}$

What is your prediction?

Games with Pure Strategies

Essentials

- To 'solve' the auctions, we need tools of non-cooperative game theory.
- Non-cooperative game theory predicts how n (rational) players decide.
 - 1-person (rational) decision theory
 - n -person (rational) decision theory
- Non-cooperative game theory is 'the language' of modern microeconomics.

Games with Pure Strategies

A (normal-form) game is a triple of sets:

- A finite set of players, $N = \{1, 2, \dots, n\}$.
- A collection of sets of **pure** strategies, $\{S_1, S_2, \dots, S_n\}$.
- A set of payoff functions, $\{v_1, v_2, \dots, v_n\}$, where $v_i : S_1 \times S_2 \times \dots \times S_n \rightarrow \mathbb{R}$ for each $i \in N$.

Finiteness

- N is finite and also each S_i is finite

Extra notation for the other players (opponents):

- $S_{-i} \equiv S_1 \times S_2 \times \dots \times S_{i-1} \times S_{i+1} \times \dots \times S_{n-1} \times S_n$
- the strategy profile of opponents: $s_{-i} \in S_{-i}$
- the strategy profile of all players: $s = (s_i, s_{-i}) = (s_1, s_2, \dots, s_n)$

Games with Pure Strategies

Pure strategies

- A *pure strategy* for player i , s_i , is a *deterministic plan* of his/her actions.
- S_i is the set of all pure strategies for player i .
- A *profile of pure strategies* $s = (s_1, s_2, \dots, s_n)$, where $s_i \in S_i$ for all i , describes a pure strategies chosen by all n players in the game.
- pure = the plan is deterministic

Static vs dynamic game

- *static* = all actions are made simultaneously
- *dynamic* = actions are made sequentially (moves), after observing some of the actions (moves) of the opponents

Games with Pure Strategies

Complete information

- In a game with *complete information*, the following components are *common knowledge*:
 - all possible actions (strategies) of all players
 - all possible outcomes
 - how each combination of actions of all players (strategy profiles) affects which outcome will materialize
 - preferences of all players over outcomes
 - the order of actions = game tree (only in a dynamic game)

Incomplete information

- In contrast, in a game with *incomplete information*, preferences of all players over outcomes are *not* common knowledge.

In this lecture, we study only *static (and finite) games with complete information* in pure strategies.

Games with Pure Strategies

Beliefs (conjectures) and best-responses

- Players can form beliefs (conjectures) about their opponents' strategies.
- A **belief** of player i is a possible profile of his opponents' strategies, $s_{-i} \in S_{-i}$.

- The strategy $s_i \in S_i$ is player i 's **best response** to his opponent's strategies $s_{-i} \in S_{-i}$ if

$$v_i(s_i, s_{-i}) \geq v_i(s'_i, s_{-i}) \quad \forall s'_i \in S_i.$$

- The **best-response correspondence** of player i selects for each $s_{-i} \in S_{-i}$ a subset $B_i(s_{-i}) \subseteq S_i$ where each strategy $s_i \in B_i(s_{-i})$ is a best response to s_{-i} .
- If the correspondence assigns only one element (i.e., if the best response strategy is unique), we speak of a **best-response function**.

Back to our example

Example (Two bidders with known asymmetric valuations, discrete bids)

- $v_1 = 10, v_2 = 5$
- integer bids: $b_i \in \{0, 1, 2, \dots\}, i = 1, 2$
- sealed-bid first-price auction

What is the best response of a player i ?

- 3 types of strategies: loss, tie, win
- let $w_i := v_i - b_{-i}$ be a *stake* of player i
- payoff for any loss strategy: 0
- payoff for tie strategy ($b_i = b_{-i}$): $\frac{1}{2}(v_i - b_{-i}) = \frac{w_i}{2}$
- payoff for the 'cheapest' win strategy ($b_i = b_{-i} + 1$): $v_i - b_{-i} - 1 = w_i - 1$

Example: Two bidders with known valuations and discrete bids

For example, what is best response of the stronger player 1 with $v_1 = 10$?

- $b_2 \leq 7$, stake $w_1 \geq 3$: **win** is best response, $w_1 - 1 > \frac{w_1}{2}$, equivalently, $\frac{w_1}{2} > 1$

$$\frac{w_1}{2} \geq \frac{3}{2} > 1$$

- $b_2 = 8$, stake $w_1 = 2$: both **tie** and **win** are best responses

$$w_1 - 1 = 1 = \frac{w_1}{2}$$

- $b_2 = 9$, stake $w_1 = 1$: only **tie** is best response

$$w_1 - 1 = 0 < \frac{1}{2} = \frac{w_1}{2}$$

- $b_2 = 10$, stake $w_1 = 0$: **tie** and **loss** are best responses

$$\frac{w_1}{2} = 0$$

- $b_2 > 10$, stake $w_1 < 0$: **loss** is best response

Knowing best responses of the stronger player, what is now your prediction?

Solution concepts

Solution = a method of analyzing games with the objective of restricting the set of all possible outcomes to those that are more 'reasonable' than others.

- a solution concept *predicts* not only the outcome but also the players' strategy profile
- prediction of a solution concept is called an *equilibrium*
- each solution concept imposes different assumptions on the players

Classic game theory derives solutions from these assumptions/axioms:

- A1 consistency/rationality of preferences
- A2 payoff-maximization
- A3 common knowledge of both rationality and payoff-maximization
- A4 correct beliefs (conjectures) about the opponents' strategies

Nash equilibrium (NE)

Nash equilibrium

- The pure-strategy profile $s^* = (s_1^*, s_2^*, \dots, s_n^*) \in S$ is a **Nash equilibrium** if s_i^* is a best response to s_{-i}^* for all $i \in N$.

TL;DR: NE is when no profitable individual deviation exists (mutual best responses).

Nash equilibrium (NE)

- Formally, NE is a solution concept defined by A1–A4.
- A4: Where do correct beliefs (conjectures) come from? How can we predict that strategic uncertainty is ‘resolved’?
 - Deductive reasoning (‘putting yourself in others shoes’) due to common knowledge of rationality and payoff-maximization (A3). (Recall no cognitive limits.)
 - Experience with past play of the opponents may form self-fulfilling beliefs.
 - Evolutionary/mass-action arguments: individuals with more successful strategies replicate faster. (In certain evolutionary games, evolutionarily stable outcomes simply correspond to Nash equilibria of baseline games.)
- Main issues with NE:
 - non-existence (too few predictions): to be addressed by mixed strategies
 - non-uniqueness (too many predictions): to be addressed by selection criteria or ‘perturbations’ to the game

Discrete Sealed-Bid Auctions with Known Asymmetric Values

Setup

Bids and values

- 2 bidders submit *integers* (discrete bids) b_1 and b_2 in a sealed-bid auction.
- If both bidders submit an identical bid, the winner is determined by toss of a coin.
- Player 1 is *stronger* (higher value/more interested) than Player 2:

$$v_1 \gg v_2 \gg 0$$

- Their valuations (v_1, v_2) are known integers.
- = a static game with complete information

Sealed-bid auctions

- *first-price auction* (FPA)
- *second-price auction* (SPA)

First-price auction

Each bid leads to either of three outcomes: win, tie, or loss.

- FPA's win payoff: $\pi_i = v_i - b_i$
- tie payoff: $\pi_i = \frac{v_i - b_i}{2}$
- loss payoff: $\pi_i = 0$

$b_i \setminus b_{-i}$	...	$v_i - 3$	$v_i - 2$	$v_i - 1$	v_i	$v_i + 1$	...
...	...	...	...	...	...	...	...
$v_i - 3$	...	$\frac{3}{2}$	0	0	0	0	...
$v_i - 2$	...	2	$\frac{2}{2}$	0	0	0	...
$v_i - 1$	...	1	1	$\frac{1}{2}$	0	0	...
v_i	...	0	0	0	0	0	...
$v_i + 1$	...	-1	-1	-1	-1	$-\frac{1}{2}$	...
...	...	...	...	...	...	...	...

Table: Row player's payoffs in FPA

- blue payoff is the diagonal (tie) payoff
- contour levels below diagonal are *horizontal* and decreasing

First-price auction

Nash equilibria in FPA

NE can be found by checking mutual best-response conditions (*necessary equilibrium conditions*) for both types of outcomes:

- (Tie, Tie)
 - If $b_1 = b_2 > v_2$ (overbidding), then the weaker player deviates to lose ($v_2 - b_2 < 0$).
 - If $b_1 = b_2 \leq v_2$, then the stronger player deviates to win ($\frac{v_1 - b_1}{2} < v_1 - b_1 - 1$.)
- (Win, Loss) $b_2 > b_1$
 - The weaker player wins.
 - Either the weaker player overbids; but then she deviates to lose.
 - Or the weaker player doesn't overbid; but then the stronger player deviates to win (see above).

First-price auction

Nash equilibria in FPA

- (Win, Loss) $b_1 > b_2$
 - C1 The weaker player doesn't (strictly) prefer to tie/win, $b_1 \geq v_2$.
 - C2 The stronger player doesn't (strictly) prefer to lose, $b_1 \leq v_1$.
 - C3 The stronger player cannot decrease her bid and still win; hence *a threat of the weak player's competing bid* must exist, $b_2 = b_1 - 1$.
 - C4 The stronger player doesn't (strictly) prefer to tie, $v_1 - b_2 - 1 \geq \frac{v_1 - b_2}{2}$

$$\frac{v_1 - b_2}{2} \geq 1$$

$$v_1 \geq b_2 + 2$$

$$v_1 \geq b_1 + 1$$

$$b_1 \leq v_1 - 1$$

- Nash equilibrium: Any $(b_1, b_1 - 1)$ such that $v_2 \leq b_1 \leq v_1 - 1$.
- In our special case with $(v_1, v_2) = (10, 5)$, the predicted bids are either $(5, 4)$, $(6, 5)$, $(7, 6)$, $(8, 7)$, or $(9, 8)$.
- ! Player 1 wins in each NE, but it is unclear on which NE the bidders coordinate.

Dominance

- Let $s_i \in S_i$ and $s'_i \in S_i, s'_i \neq s_i$ be possible strategies for player i .
- s'_i is **strictly dominated** by s_i if for *any* possible profile of the other players' strategies, $s_{-i} \in S_{-i}$, player i 's payoff from s'_i is strictly less than that from s_i :

$$v_i(s_i, s_{-i}) > v_i(s'_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}$$

- s'_i is **weakly dominated** by s_i if (i) for *any* possible profile of the other players' strategies, $s_{-i} \in S_{-i}$, player i 's payoff from s'_i is weakly less than that from s_i ,

$$v_i(s_i, s_{-i}) \geq v_i(s'_i, s_{-i}) \quad \forall s_{-i} \in S_{-i},$$

and (ii) there exists a profile of the other players' strategies, $s_{-i} \in S_{-i}$, such that player i 's payoff from s'_i is strictly less than that from s_i ,

$$\exists s_{-i} \in S_{-i} : v_i(s_i, s_{-i}) > v_i(s'_i, s_{-i}).$$

First-price auction

Strict dominance in FPA

- There is no strictly dominated strategy because for each pair (b_i, b'_i) such that $b'_i \neq b_i$, there exists a high b_{-i} such that $\pi_i(b_i, b_{-i}) = \pi_i(b'_i, b_{-i}) = 0$.

Weak dominance in FPA

- First, any excessive bid $b_i > v_i$ is weakly dominated by a *truthful* bid $b'_i = v_i$.
- Second, the truthful bid $b_i = v_i$ is weakly dominated by a bid $b'_i = v_i - 1$.
- Weakly undominated strategies are $b_i \leq v_i - 1$.

First-price auction

Nash equilibria in weakly-undominated strategies in FPA

- Weakly undominated strategies of the weaker player are $b_2 \leq v_2 - 1$.
- A *unique* NE profile in weakly-undominated strategies is

$$(b_1, b_2) = (v_2, v_2 - 1).$$

- By this prediction, the stronger player wins the item and grabs *full* surplus.
- In our special case with $(v_1, v_2) = (10, 5)$, the predicted bids are $(5, 4)$.

Second-price auction

- SPA's win payoff: $\pi_i = v_i - \min\{b_1, b_2\} = v_i - b_{-i} = w_i$
- tie payoff: $\pi_i = \frac{v_i - b_i}{2} = \frac{v_i - b_{-i}}{2} = \frac{w_i}{2}$
- loss payoff: $\pi_i = 0$

$b_i \setminus b_{-i}$	...	$v_i - 3$	$v_i - 2$	$v_i - 1$	v_i	$v_i + 1$	$v_i + 2$	...
...	...	...	...	...	...	...	...	...
$v_i - 3$	...	$\frac{3}{2}$	0	0	0	0	0	...
$v_i - 2$	...	3	$\frac{2}{2}$	0	0	0	0	...
$v_i - 1$	...	3	2	$\frac{1}{2}$	0	0	0	...
v_i	...	3	2	1	0	0	0	...
$v_i + 1$	...	3	2	1	0	$-\frac{1}{2}$	0	...
$v_i + 2$	...	3	2	1	0	-1	$-\frac{2}{2}$	...
...	...	...	...	...	...	...	...	...

Table: Row player's payoffs in SPA

- blue payoff is the diagonal (tie) payoff
- contour levels below diagonal are *vertical* and decreasing

Second-price auction

Nash equilibria in SPA

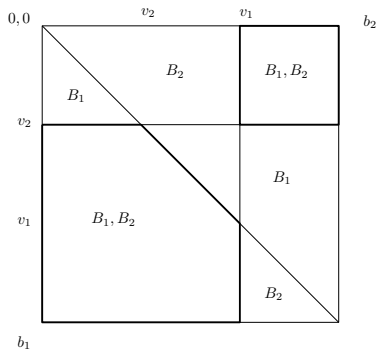
NE solved as mutual best responses. Best responses:

- If $b_{-i} < v_i$: Best response is any $b_i > b_{-i}$ (win and pay a lower bid).
- If $b_{-i} = v_i$: Best response is **any** $b_i \geq 0$.
- If $b_{-i} > v_i$: Best response is any $b_i < b_{-i}$ (lose).

Second-price auction

Nash equilibria in SPA

- The set of NE is even **richer** in SPA than in FPA.
- The stronger player 1 wins in NEs below the diagonal.
- ! The weaker player 2 wins in NEs above the diagonal.



Second-price auction

Strict dominance in SPA

- There is no strictly dominated strategy because for each pair (b_i, b'_i) such that $b'_i \neq b_i$, we have $\pi_i(b_i, v_i) = \pi_i(b'_i, v_i) = 0$.

Weak dominance in SPA

- The truthful strategy $b_i = v_i$ weakly dominates any strategy $b'_i > v_i$.
- The truthful strategy $b_i = v_i$ weakly dominates any strategy $b'_i < v_i$.
- ! In a *unique* profile of weakly undominated strategies, $(b_1, b_2) = (v_1, v_2)$, strategies are *truthful*.

Second-price auction

Nash equilibria in weakly-undominated strategies in SPA

- Any *unique* profile of weakly undominated strategies is also a NE. Here:

$$(b_1, b_2) = (v_1, v_2)$$

- In our special case with $(v_1, v_2) = (10, 5)$, the predicted bids are $(10, 5)$.
- ! Exactly like in first-price auction, if we use *NE in weakly-undominated strategies* as a solution, we predict that the stronger player wins the item, pays the price v_2 (second highest valuation), and grabs *full* surplus.

Cognitive Limits in Games

Level- k players

- Some players are strategically naïve (high costs/limits of cognition), some are strategically sophisticated (low costs/limits of cognition).
 - How to measure strategic sophistication/ability?
 - Level- k player = makes k steps of 'iterative reasoning' and believes that *all other players* are distributed between level-0 and level- $(k - 1)$ (e.g. Poisson distribution)
- = a measure of the 'depth of reasoning'
- Nash equilibrium: all players are level- ∞ players

Level- k players

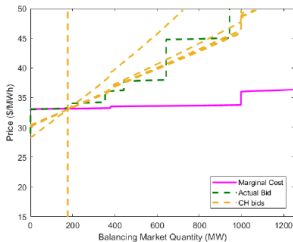
Example (Keynesian beauty contest)

- Choose $x_i \in [0, 100]$ that is closest to $p\bar{x}$, where $\bar{x}$ is the mean of x_i .
- Let $p = \frac{1}{2}$. (Keynes described $p = 1$.)
- Consider a large number (continuum) of players.
- For simplicity, use a distribution where all other players are level- $(k - 1)$ players.
- Level-0 selects (uniformly) randomly on $[0, 100]$ (non-strategic).
- Level-1 expects 50, and therefore chooses 25.
- Level-2 expects 25, and therefore chooses 12.5.
- ...
- Level- ∞ expects 0, and therefore chooses 0 (Nash equilibrium).

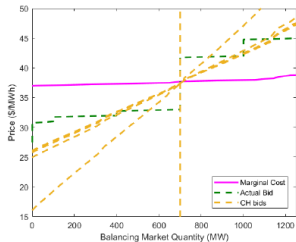
Texas spot electricity market (Hortaçsu et al., 2019)

- Texas spot electricity market (balancing auction)
- hourly bids
- bids are finalized 1 hour before the operating hour
- electricity suppliers submit bids (price-quantity pairs)
- the grid operator accepts supply bids and clears the market
- main observation: small suppliers submit overly 'vertical' bids
- Hortaçsu et al. (2019) are able to construct marginal costs; using bids and marginal costs, they identify and estimate heterogeneity in *levels of strategic sophistication* across electricity suppliers
 - larger suppliers are more sophisticated: more resources devoted to trading operations, including large trading floors, several employees with a PhD, and specially developed software applications to gather market information and compute bids
 - smaller suppliers are less sophisticated: thinly staffed, with employees who appeared to specialize in operational/engineering details of the plant rather than in trading, less time to devote to balancing market bidding

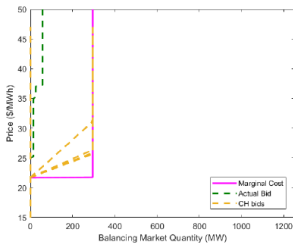
Texas spot electricity market (Hortaçsu et al., 2019)



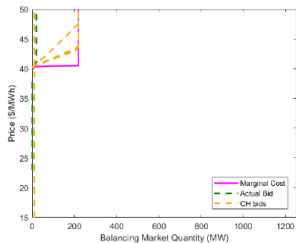
(a) Large Firm



(b) Medium Firm

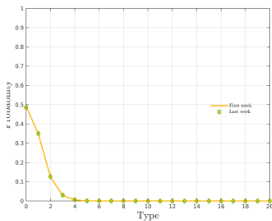


(c) Small Firm

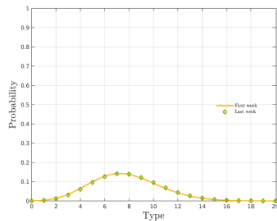


(d) Very Small Firm

Texas spot electricity market (Hortaçsu et al., 2019)



(a) Smallest Firm



(b) Largest Firm

FIGURE 8: Estimated Distributions of Types and Learning

Key lessons

- To predict bidding, game theorists recommend Nash equilibria in weakly undominated strategies.
- With known values, there is (practically) no difference between predictions for FPA and SPA. (Formally, we derived that for discrete bids and sufficiently asymmetric values of two players.)
- In both FPA and SPA, we predict that the auction is efficient (higher-valuation/stronger bidder wins) and the price is the second highest value.
- Cognitive limits (limited sophistication) matter in auctions.
- But if a *unique* NE in weakly undominated strategies exist, it should not be difficult to 'learn' it.

References

- Hortaçsu, A., Luco, F., Puller, S. L., Zhu, D. (2019). Does strategic ability affect efficiency? Evidence from electricity markets. *American Economic Review*, 109(12), 4302–4342.
- Tadelis, S. (2013). *Game Theory: An Introduction*. Princeton University Press. Chapters 3 and 5.